

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering
B.Tech II Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment - 9

1. Write C++ programs that illustrate how the following forms of inheritance are supported:
 - a) Single inheritance
 - b) Multiple inheritance
 - c) Multilevel inheritance
 - d) Hierarchical inheritance
2. To develop an application with Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club fund. Generate pay slips for the employees with their gross and net salary.
3. Write a program with the base class PLUSData has two protected member variables namely N1 and N2. Class Addition is derived from the base class AddData by using simple inheritance.
4. Write a menu-driven C++ program that performs various string manipulation operations using the std::string class. The program should display a menu to the user with options to perform the following operations:
Students have to enter their full name.
Firstname_Middlename_Surname (First letter should be capital in each word)
 - Input and Display a String
 - Find the Length of a String
 - Convert a String to Uppercase
 - Reverse a String
 - Search for a Substring
 - Replace a Word in a String
 - Exit the Program
5. Write a menu driven c++ program to perform the following operations:
 - a) Whether the entered email address by user is valid or not. Print the message accordingly.
 - b) Extract the domain name from the entered email address. And print the output.
6. Write a C++ program that takes a string as an input and check whether the string is **palindrome** or **not** ?
 - A palindrome is a word, phrase, number, or sequence of characters that reads the same backward as forward.
 - Example: madam , level.
 - a) First convert the string in lowercase.
 - b) Print “Palindrome” if true, otherwise print “Not a palindrome”.
7. Write a C++ program that takes two strings as input. Each string contains only **unique characters** (no character repeats within the same string).
 - Your task is :
 - A. Concatenate the two strings.
 - B. While concatenating, remove any characters that are common in both strings.
 - C. Print the modified concatenated string.
 - **Example:**
Input:
str1 = "abcd"
str2 = "cdef"
Output : "abef"

8. Write a C++ program that asks the user to enter a sentence. Your program should:
 - a) Count the total number of vowels (a, e, i, o, u) in the sentence.
 - b) Count the **total number of words** in the sentence.
 - c) Count the **total number of uppercase letters and lowercase letters** in the sentence.
 - d) Replace all spaces with - (hyphen) and display the modified sentence.

9. Write a C++ program to check if an entered password is valid.
 - Minimum 8 characters.
 - At least 1 uppercase letter.
 - At least 1 lowercase letter.
 - At least 1 digit.
 - At least 1 special character (@, #, \$, %, &, *).

Output should say "Strong Password" or "Weak Password" accordingly.